

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	26373	48.628167
1	[Git, Commit]	6730	12.409190
2	[Git, Change, Create]	5843	10.773684
3	[Git, Change, Delete]	4185	7.716562
4	[File, Git]	3573	6.588118
5	[Git, Change, Rename]	1699	3.132721
6	[Author, Git, Person]	999	1.842018
7	[Git, Tag]	902	1.663163
8	[Json, Key]	668	1.231700
9	[Json, Value, Scalar]	603	1.111849
10	[Committer, Git, Person]	356	0.656415
11	[NPM, Dependency]	338	0.623225
12	[Type, TS, Primitive]	293	0.540252
13	[Type, TS, Declared]	277	0.510750
14	[TS, ExternalDeclaration]	215	0.396430
15	[Type, TS, Literal]	136	0.250765
16	[Json, Value, Object]	133	0.245234
17	[Type, TS, Union]	120	0.221263
18	[Type, TS, ObjectMember]	103	0.189918
19	[NPM, Script]	91	0.167791
20	[TS, Property]	65	0.119851
21	[TS, Function]	47	0.086662
22	[Type, TS, FunctionParameter]	40	0.073754
23	[Type, Object, TS]	39	0.071911
24	[File, Directory]	34	0.062691
25	[Type, TS, Function]	34	0.062691
26	[TS, Parameter]	33	0.060847
27	[Package, File, Json, NPM]	29	0.053472
28	[TS, Variable]	24	0.044253
29	[Value, TS, Literal]	20	0.036877

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

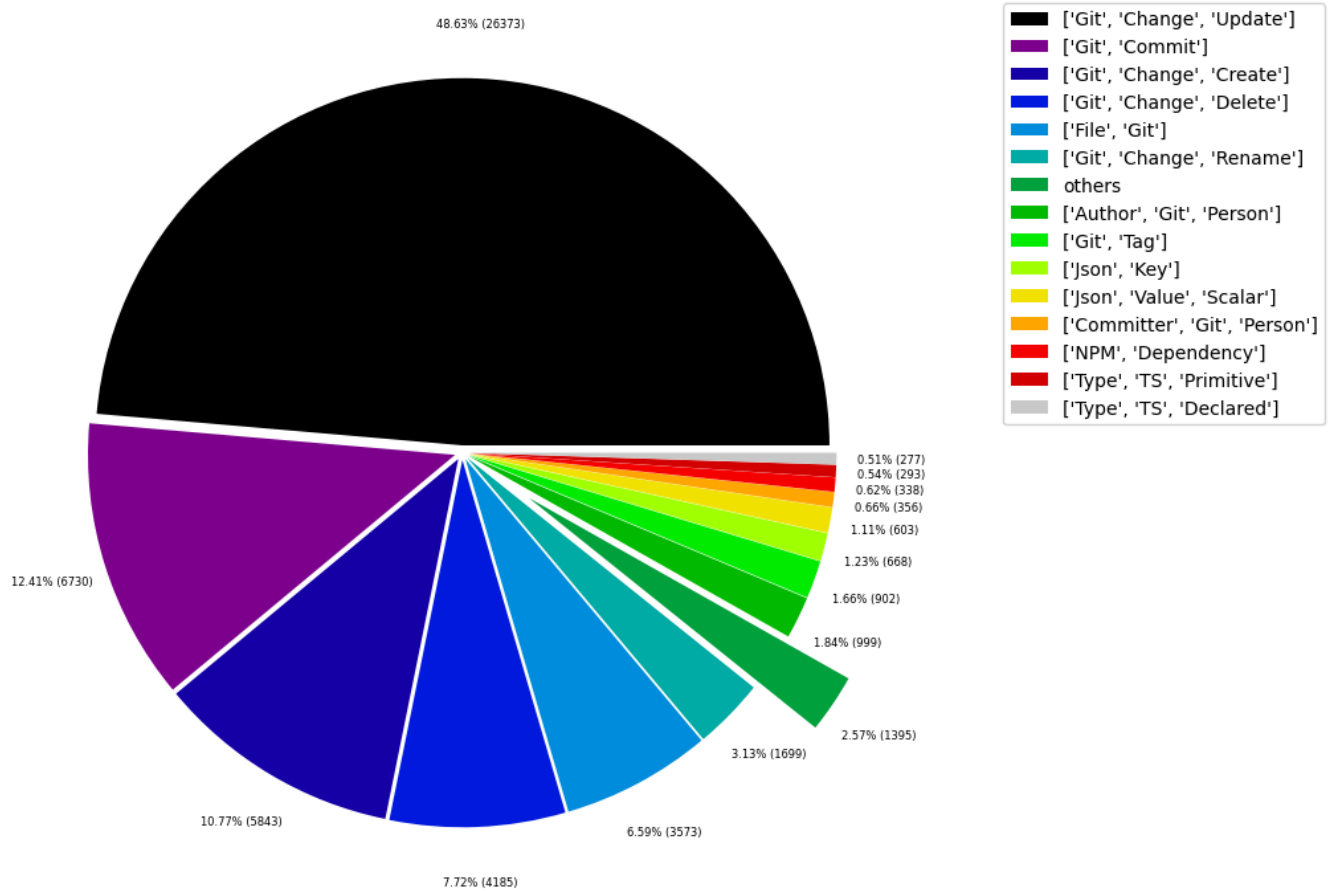


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, JQAssistant]	1	0.001844
1	[NPM, PublishConfig]	1	0.001844
2	[File, TS, Scan]	1	0.001844
3	[TS, Method]	1	0.001844
4	[Git, Branch]	1	0.001844
5	[TS, Constructor]	1	0.001844
6	[Value, TS, ObjectMember]	1	0.001844
7	[Repository, File, Git]	1	0.001844
8	[TS, Class]	1	0.001844
9	[TS, Enum]	2	0.003688
10	[Value, Object, TS]	3	0.005532
11	[Type, TS, Tuple]	3	0.005532
12	[Value, TS, Function]	4	0.007375
13	[TS, TypeParameter]	4	0.007375
14	[Value, TS, Complex]	5	0.009219
15	[Repository, NPM]	5	0.009219
16	[NPM, Engine]	6	0.011063
17	[Project, TS]	6	0.011063
18	[File, Local]	6	0.011063
19	[File, TS, Local, Module]	6	0.011063
20	[Value, TS, Call]	6	0.011063
21	[Type, TS, TypeParameterReference]	6	0.011063
22	[Value, TS, Member]	6	0.011063
23	[TS, EnumMember]	8	0.014751
24	[Type, TS, NotIdentified]	11	0.020282
25	[TS, ExternalModule]	11	0.020282
26	[Json, Value, Array]	12	0.022126
27	[Value, TS, Declared]	13	0.023970
28	[Git, Change, Copy]	15	0.027658
29	[TS, TypeAlias]	16	0.029502

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (less than 0.5% overall)

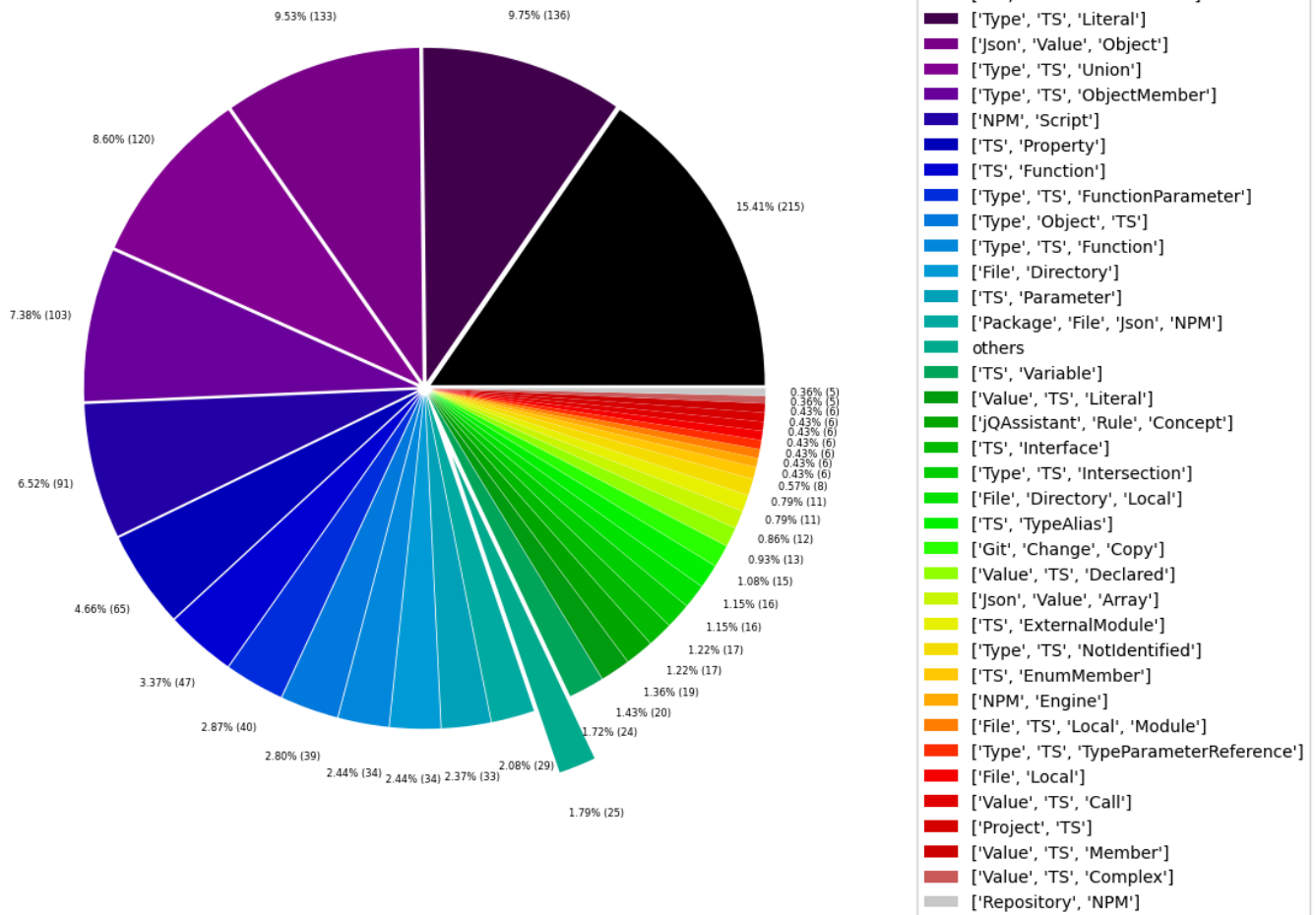


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

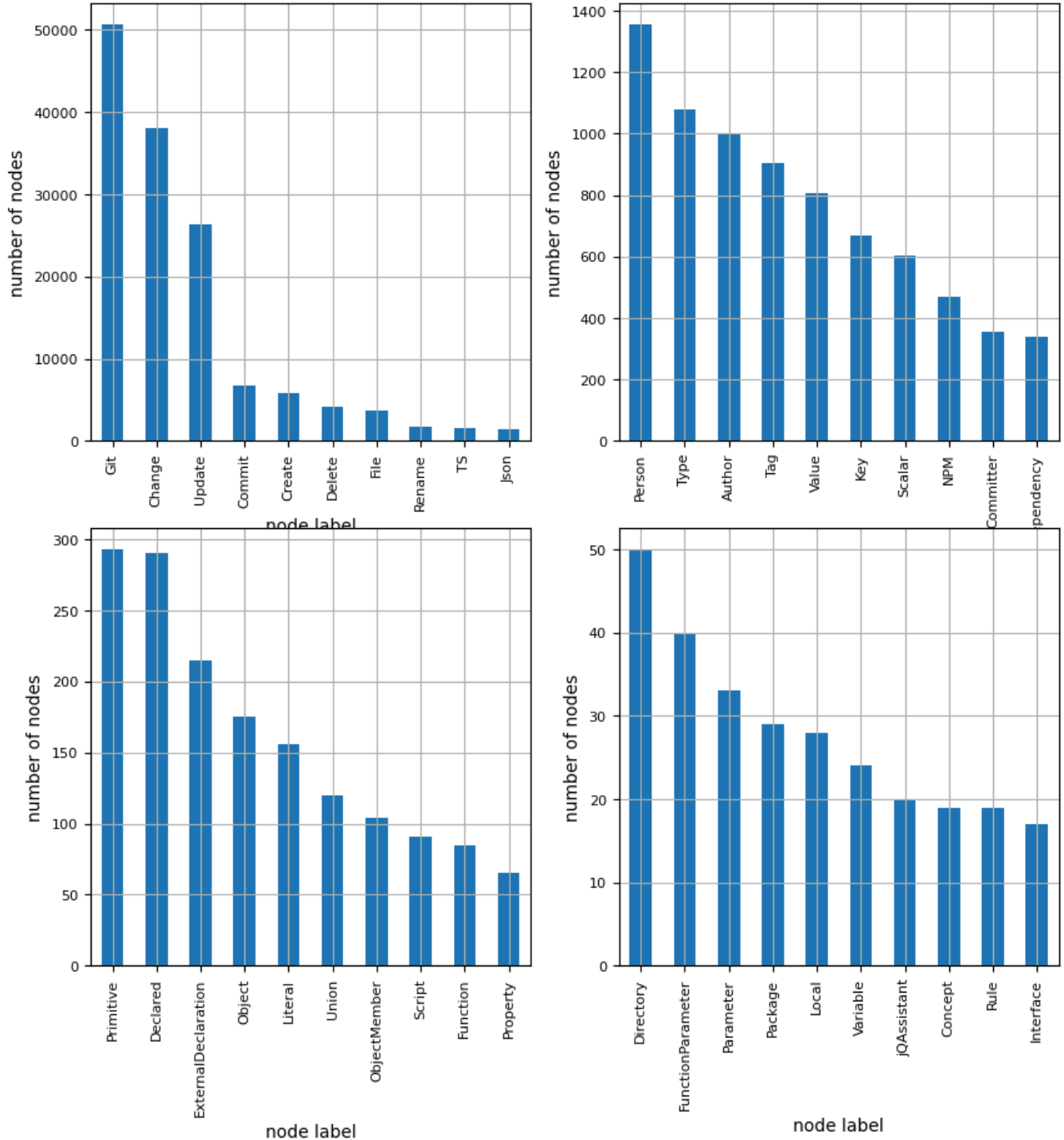
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	50677	93.441384
1	Change	38115	70.278792
2	Update	26373	48.628167
3	Commit	6730	12.409190
4	Create	5843	10.773684
5	Delete	4185	7.716562
6	File	3666	6.759597
7	Rename	1699	3.132721
8	TS	1595	2.940960
9	Json	1445	2.664380
10	Person	1355	2.498433
11	Type	1079	1.989527
12	Author	999	1.842018
13	Tag	902	1.663163
14	Value	806	1.486153
15	Key	668	1.231700
16	Scalar	603	1.111849
17	NPM	470	0.866615
18	Committer	356	0.656415
19	Dependency	338	0.623225
20	Primitive	293	0.540252
21	Declared	290	0.534720
22	ExternalDeclaration	215	0.396430
23	Object	175	0.322676
24	Literal	156	0.287642
25	Union	120	0.221263
26	ObjectMember	104	0.191762
27	Script	91	0.167791
28	Function	85	0.156728
29	Property	65	0.119851
30	Directory	50	0.092193
31	FunctionParameter	40	0.073754
32	Parameter	33	0.060847
33	Package	29	0.053472
34	Local	28	0.051628
35	Variable	24	0.044253
36	jQAssistant	20	0.036877
37	Concept	19	0.035033
38	Rule	19	0.035033
39	Interface	17	0.031346

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

<Figure size 640x480 with 0 Axes>

Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 159952

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	38115	23.829024
1	MODIFIES	38115	23.829024
2	UPDATES	26373	16.488071
3	COMMITTED	13460	8.415025
4	CREATES	7557	4.724542
5	HAS_PARENT	7502	4.690157
6	HAS_COMMIT	6730	4.207512
7	DELETES	5884	3.678604
8	HAS_FILE	3573	2.233795
9	CHANGED_TOGETHER_WITH	1801	1.125963
10	RENAMES	1699	1.062194
11	HAS_NEW_NAME	1032	0.645194
12	HAS_AUTHOR	999	0.624562
13	HAS_TAG	902	0.563919
14	ON_COMMIT	902	0.563919
15	DEPENDS_ON	887	0.554541
16	HAS_KEY	668	0.417625
17	HAS_VALUE	668	0.417625
18	CONTAINS	596	0.372612
19	HAS_COMMITTER	356	0.222567
20	OF_TYPE	339	0.211939
21	EXPORTS	309	0.193183
22	REFERENCES	197	0.123162
23	DECLARES	186	0.116285
24	DECLARES_DEV_DEPENDENCY	169	0.105657
25	DECLARES_DEPENDENCY	161	0.100655
26	HAS_MEMBER	104	0.065020
27	HAS_TYPE_ARGUMENT	94	0.058768
28	DECLARES_SCRIPT	91	0.056892
29	RETURNS	82	0.051265

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

<Figure size 640x480 with 0 Axes>

Relationship types (more than 0.5% overall)

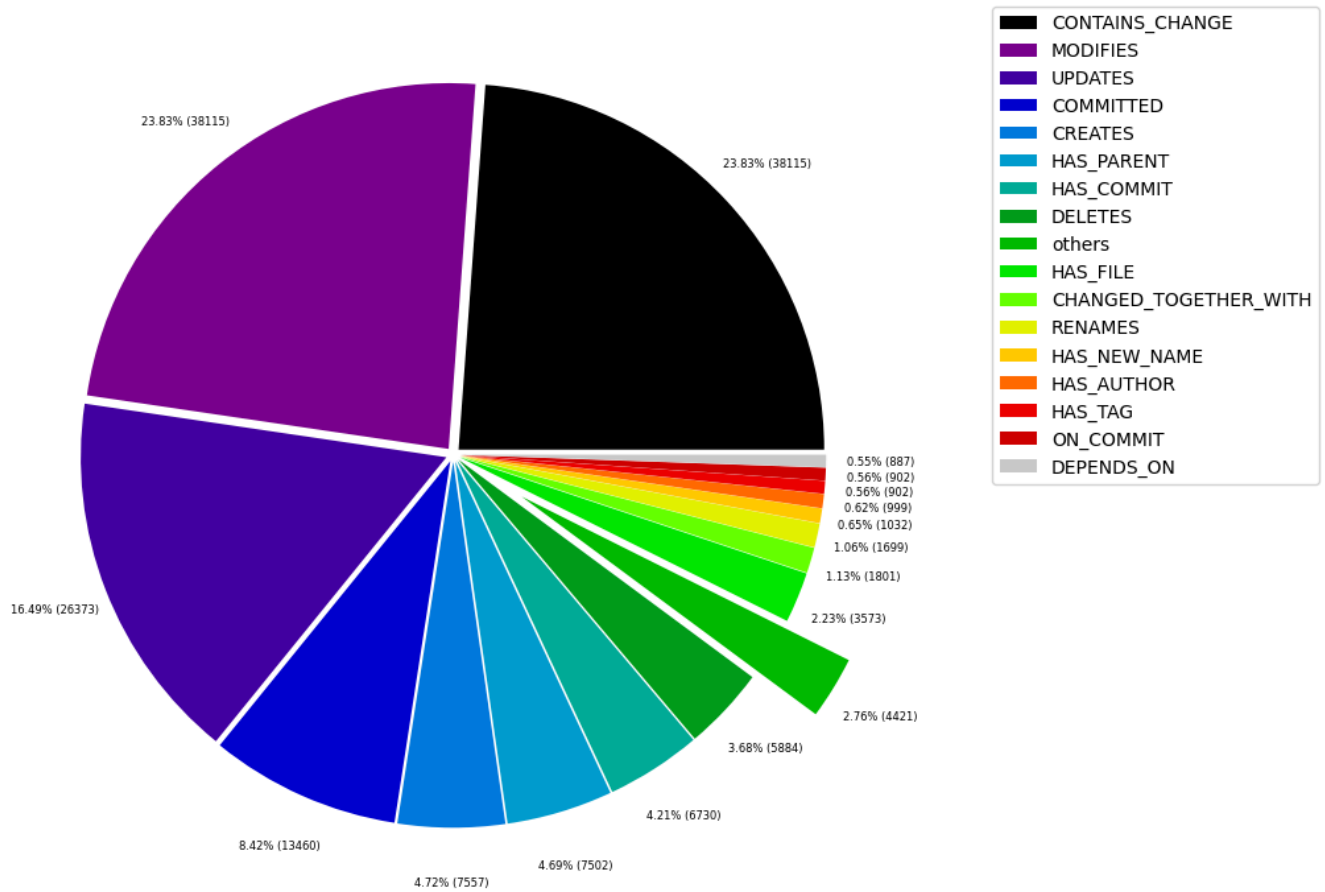


Table 2b - Lowest relationship count by type

Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

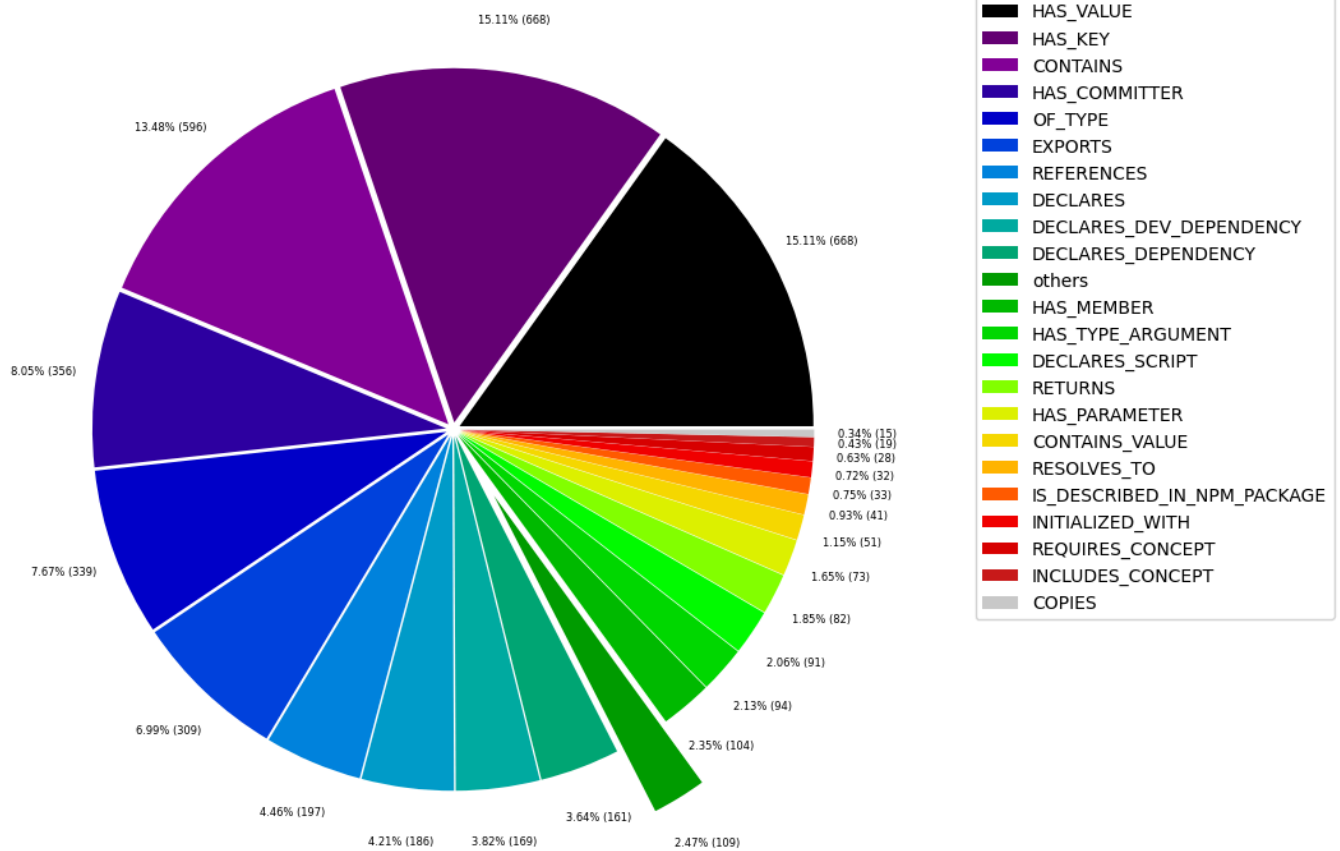
	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_BRANCH	1	0.000625
1	DECLARES_PUBLISH_CONFIG	1	0.000625
2	HAS_HEAD	2	0.001250
3	CONSTRAINED_BY	4	0.002501
4	REFERENCED_PROJECTS	5	0.003126
5	IN_REPOSITORY	5	0.003126
6	DECLARES_ENGINE	6	0.003751
7	EXTENDS	6	0.003751
8	HAS_ARGUMENT	6	0.003751
9	CALLS	6	0.003751
10	HAS_NPM_PACKAGE	6	0.003751
11	HAS_CONFIG	6	0.003751
12	PARENT	6	0.003751
13	CONTAINS_PROJECT	6	0.003751
14	MEMBER	6	0.003751
15	HAS_ROOT	6	0.003751
16	DECLARES_PEER_DEPENDENCY	8	0.005002
17	USES	11	0.006877
18	COPY_OF	12	0.007502
19	COPIES	15	0.009378
20	INCLUDES_CONCEPT	19	0.011879
21	REQUIRES_CONCEPT	28	0.017505
22	INITIALIZED_WITH	32	0.020006
23	IS_DESCRIBED_IN_NPM_PACKAGE	33	0.020631
24	RESOLVES_TO	41	0.025633
25	CONTAINS_VALUE	51	0.031885
26	HAS_PARAMETER	73	0.045639
27	RETURNS	82	0.051265
28	DECLARES_SCRIPT	91	0.056892
29	HAS_TYPE_ARGUMENT	94	0.058768

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>

Relationship types (less than 0.5% overall)



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numberOfNodesWithSameLabelsAsTarget
0	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	26373	6730	
1	[Git, Change, Update]	MODIFIES	[File, Git]	26373	26373	
2	[Git, Change, Update]	UPDATES	[File, Git]	26373	26373	
3	[Git, Commit]	HAS_PARENT	[Git, Commit]	7502	6730	
4	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	6730	1	
5	[Author, Git, Person]	COMMITTED	[Git, Commit]	6730	999	
6	[Committer, Git, Person]	COMMITTED	[Git, Commit]	6730	356	
7	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	5843	6730	
8	[Git, Change, Create]	MODIFIES	[File, Git]	5843	5843	
9	[Git, Change, Create]	CREATES	[File, Git]	5843	5843	
10	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	4185	6730	
11	[Git, Change, Delete]	DELETES	[File, Git]	4185	4185	
12	[Git, Change, Delete]	MODIFIES	[File, Git]	4185	4185	
13	[Repository, File, Git]	HAS_FILE	[File, Git]	3573	1	
14	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	1699	6730	
15	[Git, Change, Rename]	MODIFIES	[File, Git]	1699	1699	
16	[Git, Change, Rename]	DELETES	[File, Git]	1699	1699	
17	[Git, Change, Rename]	CREATES	[File, Git]	1699	1699	
18	[Git, Change, Rename]	RENAMES	[File, Git]	1699	1699	
19	[File, Git]	CHANGED_TOGETHER_WITH	[File, Git]	1648	3573	
20	[File, Git]	HAS_NEW_NAME	[File, Git]	1032	3573	
21	[Repository, File, Git]	HAS_AUTHOR	[Author, Git, Person]	999	1	
22	[Repository, File, Git]	HAS_TAG	[Git, Tag]	902	1	
23	[Git, Tag]	ON_COMMIT	[Git, Commit]	902	902	
24	[Json, Value, Object]	HAS_KEY	[Json, Key]	668	133	
25	[Json, Key]	HAS_VALUE	[Json, Value, Scalar]	552	668	
26	[Repository, File, Git]	HAS_COMMITTER	[Committer, Git, Person]	356	1	
27	[TS, Function]	DEPENDS_ON	[TS, ExternalDeclaration]	293	47	
28	[File, TS, Local, Module]	DEPENDS_ON	[TS, ExternalDeclaration]	236	6	
29	[TS, ExternalModule]	EXPORTS	[TS, ExternalDeclaration]	215	11	

Graph Density

total_number_of_nodes (vertices): 54234

total_number_of_relationships (edges): 159952

-> total directed graph density: 5.4381904024056975e-05

-> total directed graph density in percent: 0.005438190402405697